

ITA0506 - Computer Vision

1.

a. Main objectives of computer vision:-
Computer vision is an AI agent which is used to interpret, scan, train the image and enhance image quality using a computer.

b. Real-world applications.

- Automatic Vehicle for driving.
- X-ray Scannings.
- Object/fraud object detection.

2.

a. Feature Extraction:-

The object is detected using the sensors and enhance the contrast and Noise reductions are reduced it is known as feature Extraction and identifies distinct components.

b. The object recognition improves computer vision systems using the pixels and the pixels is brighten by using Noise reduction then the object is done with quality.

3.

a. low-level image processing:-

It mainly focuses on basic operations like Noise Reduction, edge detection and image sharpening - image enhancement and contrast enhancement.

b) Mid-level vs high level processing:-

- Mid level mainly focus on edge detection, edge sharpening and edge of the object and corners of the object.
- High-level mainly focuses on detection of the object and recognition of the object.

- 4.
- Pixel:- A pixel is a the small individual distinct in a digital image, possess the specific photons colour intensity value.
 - Effect of resolution image quality.
Higher resolution means a greater number of pixels which captured by the sensors.
- 5.
- Role of an Image sensor:-
Image sensor are like CCD and CMOS capturing light from the real-world scene and converts into electrical signals.
 - Two acquisition devices:-
1) digital camera
2) Smartphone camera.
- 6.
- Sampling:- Sampling is the digitisation of the spatial co-ordinates of an image, dividing the continuous real-world scene into a discrete grid of points.
 - Purpose of quantization.
Quantization maps the continuous infinite range of the intensity values of the samples into a finite, discrete set.
- 7.
- perspective projection:-
It is a mathematical modeling how 3D objects in the physical world are projected as 2D image onto a camera plane.
 - Role of Illumination:- Illumination is a source of light interacting with the physical properties.

Define an
Image
Two
1) U
2)

Define image processing:-

Image processing is the process of manipulating digital image to enhance its quality or extract its basic features

b. Two main differences:-

- 1) Using machine learning techniques
- 2) Using visual quality.

9. Real-world application

a. Health-care application

In Radiology, computer vision analyzes the MRI, scans the body and automatically detects the unwanted tumors.

b) autonomous vehicles:-

The vehicles which can drive without and human being help. It detects the road and persons on the Road uses AI to detect the objects.

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- a. An image is stored and processed by the computer as a matrix of these numerical values by assessing the spatial arrangement and Range of intensity

Influence and image resolution

- b. Together, they determine the amount of visible detail and the digital size of file.